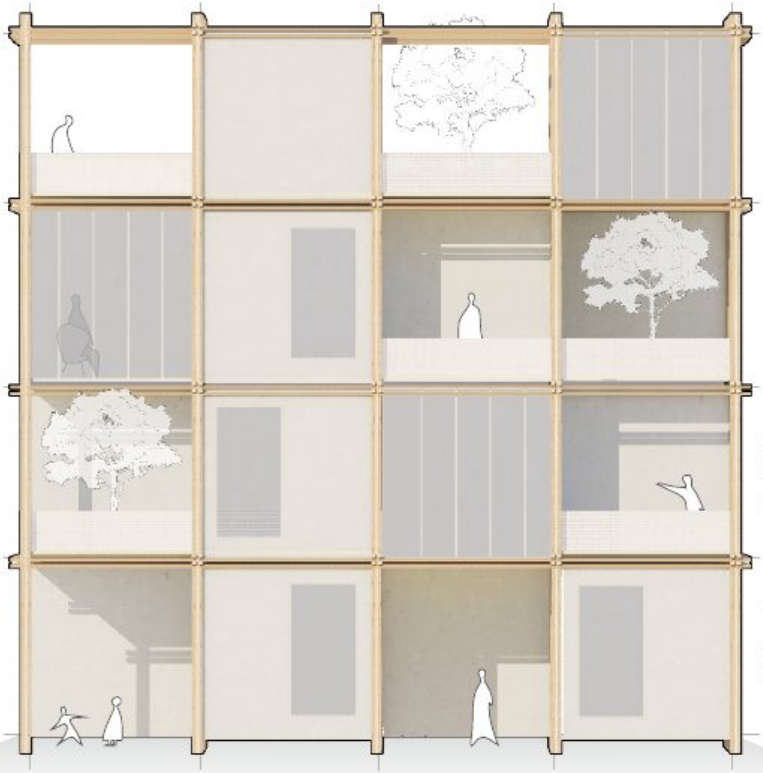
02 "TOD" in 2046

The Noise Problem leads to the Exploration of the Ideal Future Residential Ensemble



Location: Baiyun District, Guangzhou, China
Date: Spring Term 2023-2024
Instructor: Prof. Yang Zhao
Acadamic Work: Architectural Design V, Individual Work

A happened conversation learned that a friend lives in this neighborhood, gave me a lot of feedback on the noise problem, which I used as a subject to start the program.

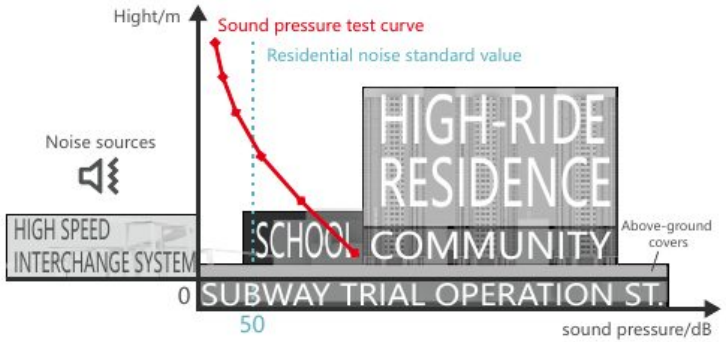
The project is located in Baiyun District, Tinggang Station subway superstructure real estate projects, I analyzed the internal composition of the plot coincidentally.

An Opportunity

During a greeting with a friend, he didn't seem to be in a very good state, and I learned that he had been suffering from **the noise of the highway** next to his home and suffered from **frequent insomnia**. This has aroused my interest and thoughts, and I hope to be able to help.

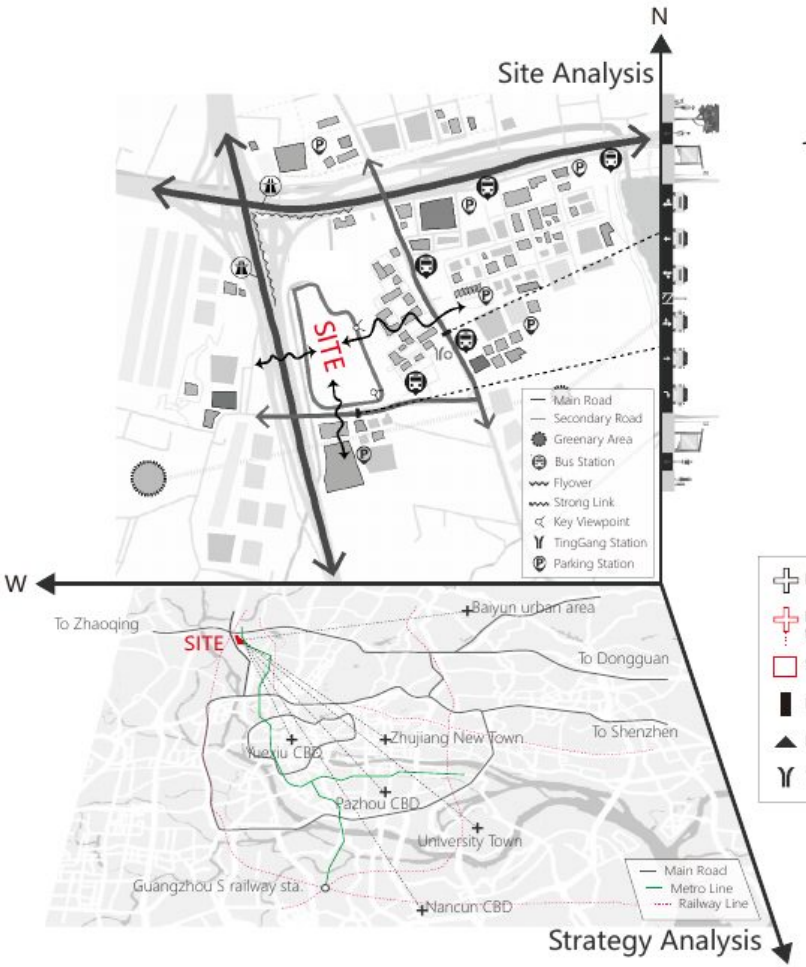


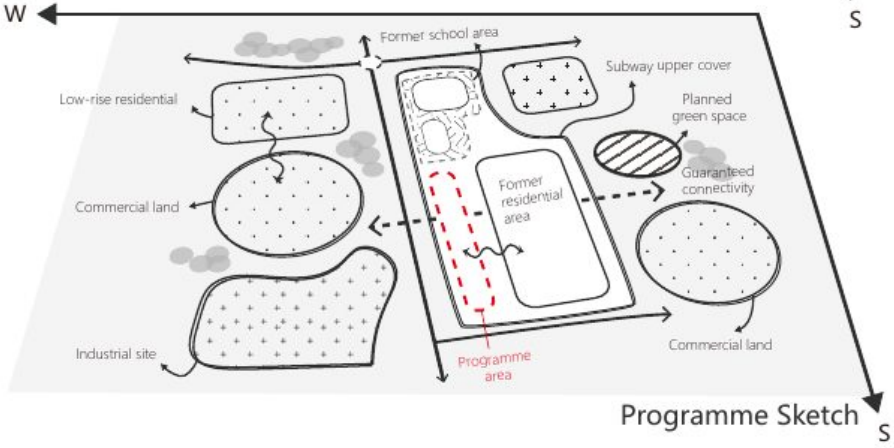
On-site Noise Survey Data

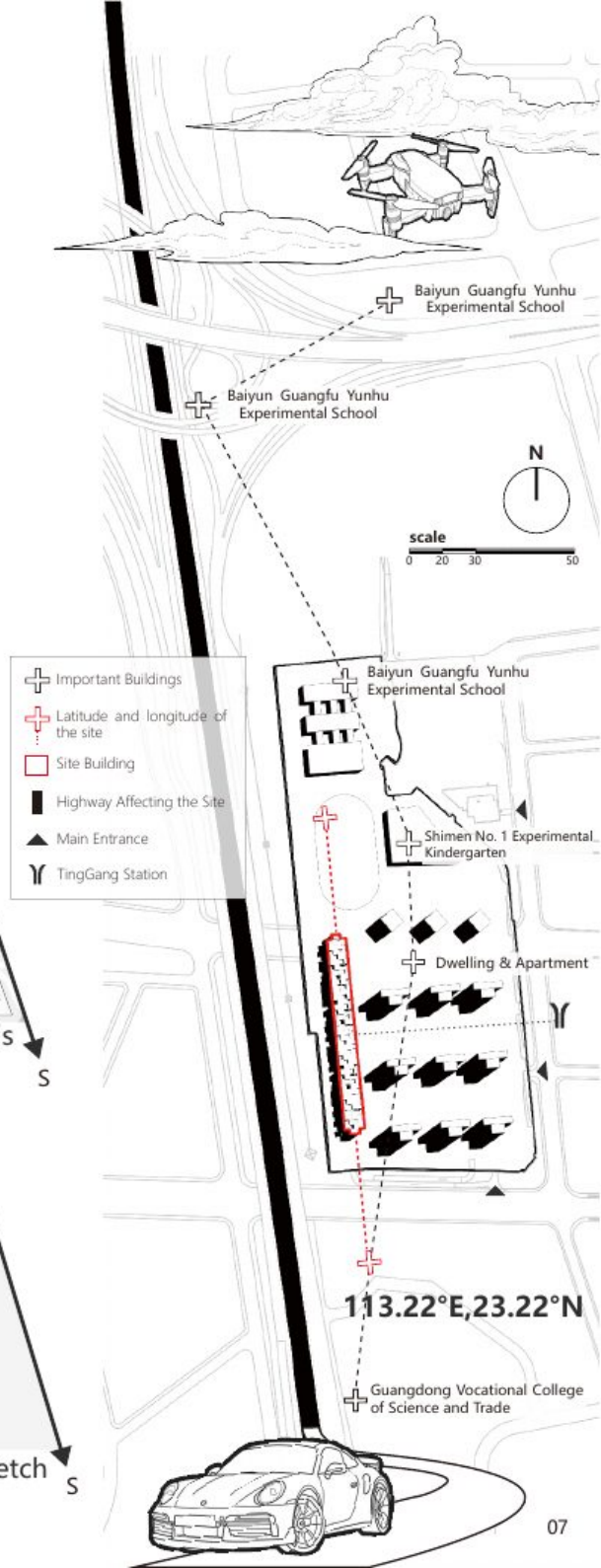


According to the investigation, it was preliminarily judged that the noise in the **low-rise residential** and **school areas** seriously exceeded the standard value, which had a great impact on daily use.

Preliminary Research







Logical Deduction

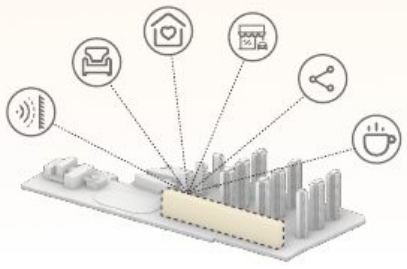
derive spatial logic based on community needs

Vertical exchange space
Breaking up the floor slabs between floors to form a joint vertical green space.

Semi-outdoor space
Open up the view between the floors, place greenery, create a courtyard, and improve the suitability of living.

Core space
Vertical traffic connects these combined block spaces to improve accessibility.

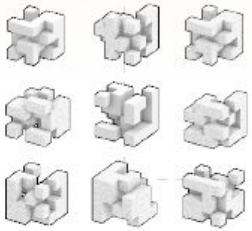
Aggregate Space
Blocks gathered together can form a large shared space, breaking down the barriers between layers.



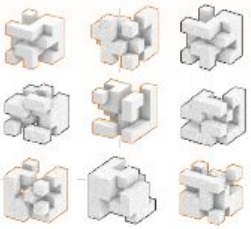
Objectives & Actions

The site needed to create a **mixed-use** building that not only solves the problem of sound insulation, but also meets the needs of commercial, apartment, community service, and shared leisure functions.

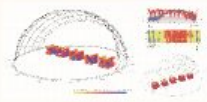
1. Randomly generate the 9x9 boxes.



2. Evaluate the possibilities of each square and select possible boxes.



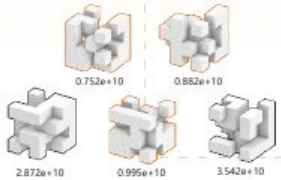
3. Evaluate its insolation conditions,Pick the top 5 boxes.



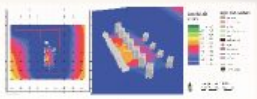
4. Verify the results of the assessment.



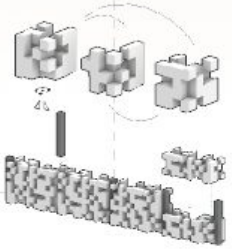
5. Extract the underside that mainly produces the echo.



6. calculate the area sum,Pick the top 3 boxes.



8. Noise evaluation of the results.



7. Permutations and combinations to form a staircase.





Derived from the isolation of noise, this conglomerate combining **commercial, residential, leisure, community services**, children's activities and other functional needs will become an **ideal utopia** for people in a high-density city.

